MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: PLUG, RIGHT ANGLE, DECADE INCREMENT 20 THRU 110 AND 128 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.050 SPACING)

This specification is approved for use by Rome Air Development Center, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the connectors described herein shall consist of this document and the latest issue of Specification MIL-C-55302.

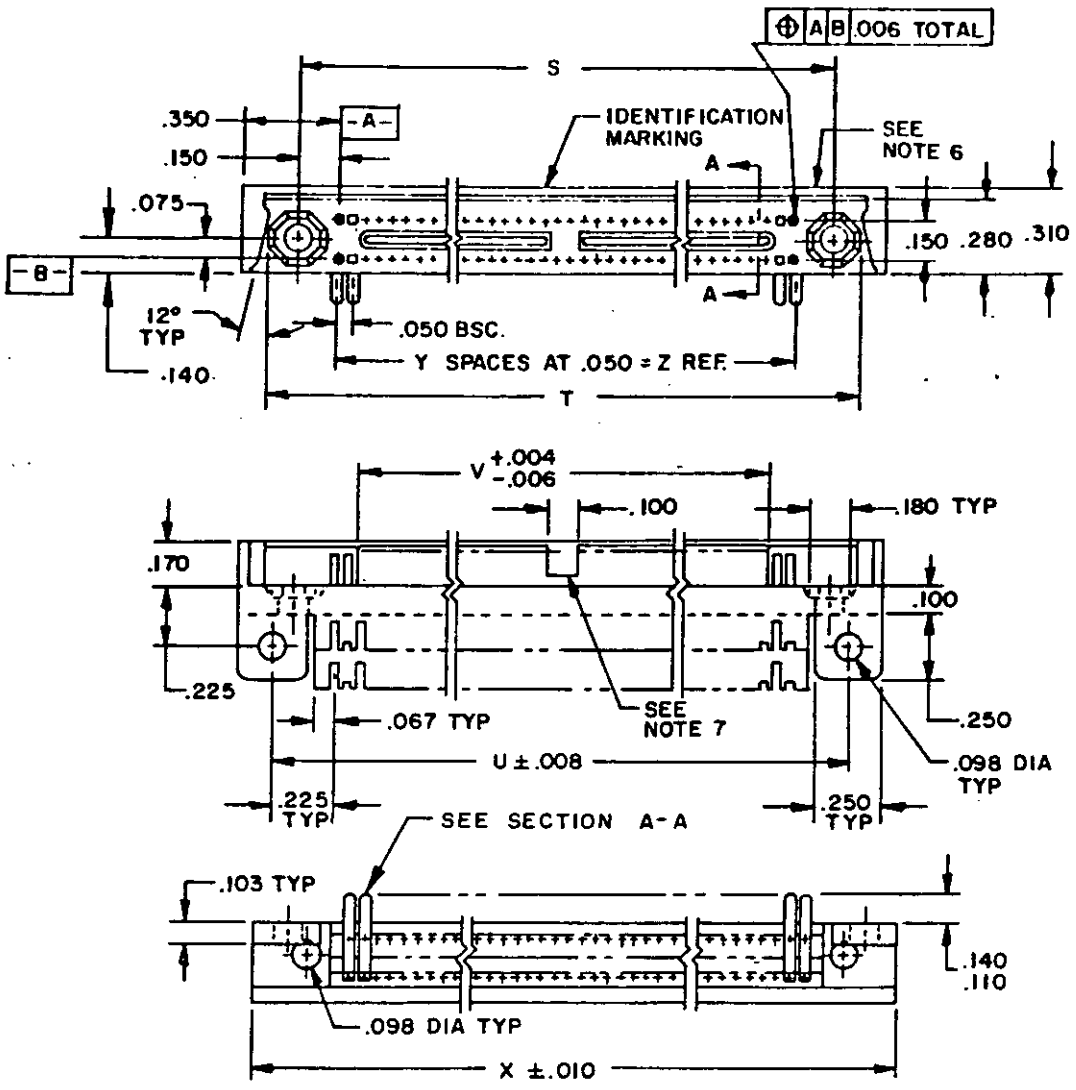
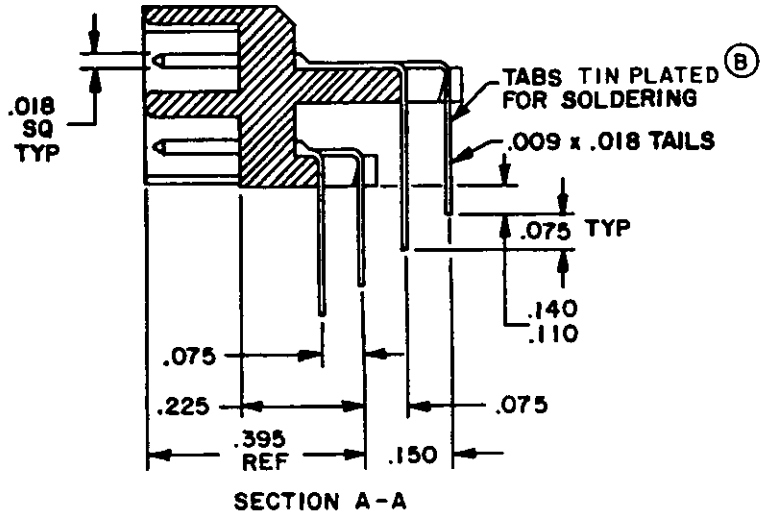


FIGURE 1. Connectors, plug .050 (1.27 mm) spacing.

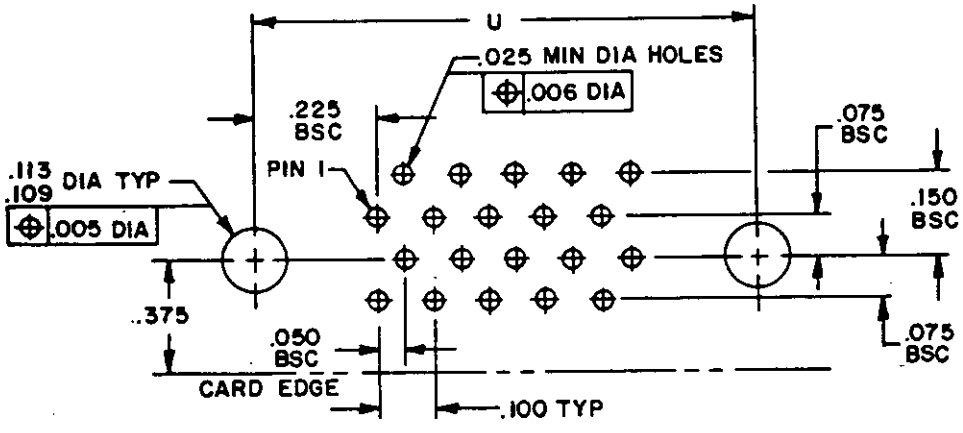
(B) denotes changes

FSC 5935

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SECTION A-A



RECOMMENDED PRINTED WIRING BOARD LAYOUT Ⓑ

INCHES	MM	INCHES	MM	INCHES	MM
.004	.10	.075	1.90	.170	4.32
.005	.13	.098	2.49	.180	4.57
.006	.15	.100	2.54	.225	5.72
.008	.20	.103	2.62	.250	6.35
.009	.23	.109	2.77	.280	7.11
.010	.25	.110	2.79	.310	7.87
.018	.46	.113	2.87	.350	8.89
.050	1.27	.140	3.56	.375	9.52
.067	1.70	.150	3.81	.395	10.03

FIGURE 1. Connectors, plug .050 (1.27 mm) spacing - Continued.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm) and ±2° on angles.
4. These connectors mate with connectors specified in MIL-C-55302/119(USAF).
5. Pad (s) suitable for printed wiring board support are optional. Dimensions and locations are optional.
6. Numbers indicating end positions stamped this side only.
7. Except 20 thru 50 position.

FIGURE 1. Connector, plug .050 (1.27 mm) spacing - Continued.

TABLE I. Dash numbers and dimensions.

Dash number	No. of contacts	Dimensions							Contact identification numbers
		S	T	U	V	X	Y	Z	
01	20	.750 (19.05)	.970 (24.64)	.900 (22.86)	.290 (7.37)	1.150 (29.21)	9	.450 (11.43)	1-10
02	30	1.000 (25.40)	1.220 (30.99)	1.150 (29.21)	.540 (13.72)	1.400 (35.56)	14	.700 (17.78)	1-15
03	40	1.250 (31.75)	1.470 (37.34)	1.400 (35.56)	.790 (20.07)	1.650 (41.91)	19	.950 (24.13)	1-20
04	50	1.500 (38.10)	1.720 (43.69)	1.650 (41.91)	1.040 (26.42)	1.900 (48.26)	24	1.200 (30.48)	1-25
05	60	1.750 (44.45)	1.970 (50.04)	1.900 (48.26)	1.290 (32.77)	2.150 (54.61)	29	1.450 (36.83)	1-30
06	70	2.000 (50.80)	2.220 (56.39)	2.150 (54.61)	1.540 (39.12)	2.400 (60.96)	34	1.700 (43.18)	1-35
07	80	2.250 (57.15)	2.470 (62.74)	2.400 (60.96)	1.790 (45.47)	2.650 (67.31)	39	1.950 (49.53)	1-40
08	90	2.500 (63.50)	2.720 (69.09)	2.650 (67.31)	2.040 (51.82)	2.900 (73.66)	44	2.200 (55.88)	1-45
09	100	2.750 (69.85)	2.970 (75.44)	2.900 (73.66)	2.290 (58.17)	3.150 (80.01)	49	2.450 (62.23)	1-50
10	110	3.000 (76.20)	3.220 (81.79)	3.150 (80.01)	2.540 (64.52)	3.400 (86.36)	54	2.700 (68.58)	1-55
11	128	3.450 (87.63)	3.670 (93.22)	3.600 (91.44)	2.990 (75.95)	3.850 (97.79)	63	3.150 (80.01)	1-64

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REQUIREMENTS:

Design and construction:
Dimensions and configurations: See figure 1.
Material:
Body: Type GPT-30F per MIL-M-24519.
(B) Plating (contact): Gold per MIL-G-45204, type II, grade C, class I over nickel per QQ-N-290, class 2, 30 to 150 microinches on the contact engagement area for .100 inches minimum length. The termination area plating shall be tin lead per MIL-T-10727, class I over nickel per QQ-N-290, class 2, 30 to 150 microinches, for .300 inches minimum length.
(B) Contact identification: See figure 1.
Contact rating: 1.5 amperes per contact, maximum.
Operating temperature: -65° to 125°C.
Keying: See MIL-C-55302/31.

Mating and unmating: The maximum mating force in pounds shall be the number of contacts multiplied by .30 and the withdrawal force in pounds shall be a minimum of .03 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance: The average contact resistance of all contacts measured shall not exceed .025 ohm, and no individual contact pair shall have a resistance exceeding .030 ohm.

(B) Contact retention: .75 pound, minimum.

Dielectric withstanding voltage:
Sea level: 600 volts rms.
High altitude: 150 volts rms.

(B) Solderability: Each connector shall be subjected to method 208 of MIL-STD-202 (Solder termination only) (Applicable to qualification and retention of qualification only).

Part number: M55302/117 - (dash number from table I).

Patent number 3,404,367. The Government has a royalty free license under this patent for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government.

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MIL-C-55302/117B (USAF)

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